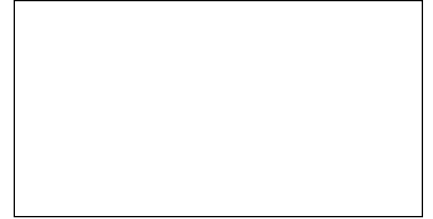




2007 - msc entrance exam for computer science aau

Computer_Security (Addis Ababa University)



**Department of Computer Science
Faculty of Informatics
Addis Ababa University**

Entrance Examination for the M. Sc. Program in Computer Science

Date: July 13, 2007
Time allowed: 3 hours

Name: _____

Instructions

1. Read the instructions and the questions carefully.
2. Write your name **only** on this page. Writing your name on the other pages will automatically disqualify you from this exam.
3. Answer all questions eligibly on the spaces provided on the question papers. Please make sure that sentences are short, and grammatically and logically correct. You can use the back side of each paper when space is not sufficient. In such a case, please indicate so.
4. This examination booklet consists of **11** pages (excluding this cover page). Make sure that the question papers are numbered consecutively (at the bottom of each page) and that no question papers are missing or replicated.
5. Cheating is strictly prohibited. Any person found cheating will be automatically disqualified from the exam.
6. Use only the rough papers provided. Using other unauthorized papers will be considered as cheating.
7. Switch **off** your mobile phones. You are not allowed to use a mobile for **any** purpose.
8. You are not allowed to use calculators or any other electronic device.

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Part	Part I (35)	Part II (25)	Part III (20)	Part IV (20)	Total (100)
Marks					

Part I: Programming, Data Structures, and Software Engineering (35 points)

1.1 a. What is a software development process model? (2 points)

the process by which an organization develops software.

b. Explain in detail the Rapid Application development model. (3 points)

The Rapid Application Development (or RAD) model is **based on prototyping and iterative model with no (or less) specific planning**. In general, RAD approach to software development means **putting lesser emphasis on planning tasks and more emphasis on development and coming up with a prototype**.

1.2 a. What is software risk? (2 points)

a potential problem that may or may not occur in the future.

the probability of occurrence for uncertain events and their potential for loss within an organization.

There are three main classifications of risks which can affect a software project:

Project risks.

Technical risks.

Business risks.

b. Discuss software process risk. (3 points)

12 risks in software development

- Code issues. One significant risk involved with software development is poor quality code. ...
- Aggressive deadlines. Sometimes, software development projects have tight deadlines. ...
- Unmet expectations. ...
- Low productivity. ...
- Budget issues. ...
- Poor risk management. ...
- Inadequate project management. ...
- Scope creep

1.3 a. What is the difference between in-fix notation and post-fix notation? (1 point)

Infix expression is an expression in which the operator is in the middle of operands, like operand operator operand.

Postfix expression is an expression in which the operator is after operands, like operand operator. Postfix expressions are easily computed by the system but are not human readable

b. Convert the expression $A + (B * C - D/E * F) * H$ into post-fix notation. (3 points)

1.4 Given a binary tree BT, write an algorithm (using pointer implementation) that counts the number of

```
1 public class CountTreeNodes {
2     static class Node{
3         Node left;
4         Node right;
5         int data;
6
7         public Node(int data) {
8             this.left = null;
9             this.right = null;
10            this.data = data;
11        }
12    }
13
14    public int countNode(Node root){
15
16        //base case
17        if(root==null)
18            return 0;
19
20        //recursive call to left child and right child and
21        // add the result of these with 1 ( 1 for counting the root)
22        return 1 + countNode(root.left) + countNode(root.right);
23    }
24
25    public static void main(String[] args) {
26        CountTreeNodes c = new CountTreeNodes();
27        Node root = new Node(1);
28        root.left = new Node(2);
29        root.right = new Node(3);
30        root.left.left = new Node(4);
31        root.left.right = new Node(5);
32
33        System.out.println("Number of nodes in given binary tree: " + c.countNode(root));
34    }
35 }
```

nodes.

(5 points)

- 1.5 Rewrite the following recursive algorithm using iterative methods using a programming language of your choice. You may introduce new variables as required. (5 points)

```
int f(int num)
Begin
/* num > 0 */
  if( num <= 2 ) then
    return 1;
  else
    return f(num - 1) + f(num - 2);
  end if
End
```

- 1.6 Mention at least three points in which Object Oriented programming is advantageous than structured programming. (3 points)

allows you to control the complexity.

It is less difficult to modify object-oriented programs and reuse code as compared to structured programs.

It provides more flexibility and abstraction

it gives more importance to data.

- 1.7 Write three benefits of modular programming? (3 points)

a general programming concept. It involves separating a program's functions into independent pieces or building blocks, each containing all the parts needed to execute a single aspect of the functionality

- Code is easier to read. ...
- Code is easier to test. ...
- Easily find things later. ...
- Reusability without bloat. ...
- Single source for faster fixes. ...
- Easier, lower risk updates. ...
- Easier refactoring. ...
- Easier to collaborate.

1.8 Write the implementation of a simple sorting algorithm with a programming language of your choice to order a list having n integer values in descending order. (5 points)

```
1.  public class SortDsc {
2.      public static void main(String[] args) {
3.          //Initialize array
4.          int [] arr = new int [] {5, 2, 8, 7, 1};
5.          int temp = 0;
6.          //Displaying elements of original array
7.          System.out.println("Elements of original array: ");
8.          for (int i = 0; i < arr.length; i++) {
9.              System.out.print(arr[i] + " ");
10.         //Sort the array in descending order
11.         for (int i = 0; i < arr.length; i++) {
12.             for (int j = i+1; j < arr.length; j++) {
13.                 if(arr[i] < arr[j]) {
14.                     temp = arr[i];
15.                     arr[i] = arr[j];
16.                     arr[j] = temp;
17.                 }
18.             } }
19.         System.out.println();
20.         //Displaying elements of array after sorting
21.         System.out.println("Elements of array sorted in descending order: ");
22.         for (int i = 0; i < arr.length; i++) {
23.             System.out.print(arr[i] + " ");
24.         }
25.     }
26. }
27. Elements of original array:
28. 5 2 8 7 1
```


29. Elements of array sorted in descending order:

30.8 7 5 2 1

Part II: Computer Architecture and Operating Systems (25 points)

2.1 In the 16-bit FLAGS register, 9 of the 16-bits are used as flags. What are the two groups in which these flag bits are grouped? Describe the purpose of each group. (4 points)

The Status Flags, and the Control Flags.

Status flags **enable an instruction to act based on the result of a previous instruction.**

(b) Control Flags – The control flags **enable or disable certain operations of the microprocessor**

2.2 Compare CISC and RISC in light of emphasis on the following points? (5 points)

Point of Emphasis	RISC	CISC
Program code size and clock cycle needed	Low cycles per second with large code sizes	Small code sizes with high cycles per second
	Single clock cycle instructions	Instructions can take several clock cycles
“LOAD” and “STORE” instructions	Small number of fixed length instructions	Large number of instructions
Transistors much needed for	Less instruction ,less transistor	Larg instruction more transistor
	Heavy use of RAM	More efficient use of RAM
Hardware or Software emphasis	Emphasis on software	Emphasis on hardware
Instructions’ complexity	Simple, standardised instructions	Complex, variable-length instructions

2.3 In von Neumann architecture, the scheme of segment plus offset allows a program code or data to be relocatable. Describe in your words what relocation is and how it is done. (4 points)

Relocation is **the process of assigning load addresses for position-dependent code and data of a program and adjusting the code and data to reflect the assigned addresses**

2.4 In most modern operating systems, making use of threads in a single process is common. Write two basic reasons (advantages) for using threads. (2 points)

Threads in a process **can execute different parts of the program code at the same time**

- Improved throughput. ...
- Simultaneous and fully symmetric use of multiple processors for computation and I/O.
- Superior application responsiveness. ...
- Improved server responsiveness. ...
- Minimized system resource usage. ...
- Program structure simplification. ...
- Better communication.

2.5 Direct Memory Access (DMA) is commonly available in general-purpose modern computer systems. What are the disadvantages of a system without a DMA? (2 points)

enables a device to transfer data by exposing the CPU to a work overload

The main disadvantages of DMA are the **additional cost of the hardware** and the added complexity of the software

2.6 What are the four types of files supported by the Unix operating system? (2 points)

1. Ordinary Files
2. Directories
3. Special Files
4. Pipes
5. Sockets
6. Symbolic Links

2.7 Paging and segmentation are two common methods of implementing virtual memory. Unlike pages, segments do not have fixed size and have independent linear address space. Give two advantages of segmentation over paging. (3 points)

Segment tables are mapped to page tables, and page tables are mapped to individual pages within a segment. Advantages include **less memory usage, more flexibility on page sizes, simplified memory allocation, and an additional level of data access security over paging**. The process does not cause external fragmentation

2.8 What is the Master Boot Record (MBR) in a file System? What happens to the system if the MBR is destroyed, say by a virus? (3 points)

The Master Boot Record (MBR) is **the information in the first sector of a hard disk or a removable drive**. As a result, when the MBR gets damaged, **the operating system simply won't load**.

Part III: Database Management Systems (20 points)

3.1 Consider the following six relations for an order processing database application in a company:

CUSTOMER(CustNo, Cname, City)

ORDER(OrderNo, Odate, CustNo, Ord_Amt)

ORDER_ITEM(OrderNo, ItemNo, Qty)

ITEM(ItemNo, Unit_price)

SHIPMENT(OrderNo, WarehouseNo, Ship_date)

WAREHOUSE(WarehouseNo, City)

'Cname' refers to Customer name, 'Ord_Amt' refers to total amount of an order in Birr; 'Odate' is the date the order was placed; 'Ship_date' is the date an order is shipped from the warehouse. Assume that an order can be shipped from several warehouses.

Write SQL statements for the following queries:

- a. List the **OrderNo** and **Ship_date** for all orders shipped from Warehouse number '**W2**'. (1 point)

answer

```
select ORDER.OrderNo,SHIPMENT.Ship_date FROM ORDER INNER JOIN SHIPMENT WHERE  
WAREHOUSE.WarehouseNo='W2';
```

- b. List the Warehouse information from which the Customer named '**Yonas**' was supplied his orders by producing a listing: **OrderNo, WarehouseNo, Warehouse_City**. (1.5 points)

- c. Produce a listing: **CUSTNAME**, **NO_OF_ORDERS**, **AVG_ORDER_AMT**, where the **NO_OF_ORDERS** is the total number of orders by the customer and the **AVG_ORDER_AMT** is the average order amount for that customer. (3 points)
- d. List the orders that were not shipped within 30 days of ordering. (1.5 points)
- e. List the **OrderNo** for orders that were shipped from **all** warehouses that the company has in '**Addis Ababa**'. (3 points)

3.2 What is the difference between logical data independence and physical data independence? (3 points)

answer

Logical data is data about database, that is, it stores information about how data is managed inside. For example, a table *relation* stored in the database and all its constraints, applied on that relation.

Logical data independence is a kind of mechanism, which liberalizes itself from actual data stored on the disk. If we do some changes on table format, it should not change the data residing on the disk.

Physical data independence All the schemas are logical, and the actual data is stored in bit format on the disk. It is the power to change the physical data without impacting the schema or logical data.

For example, in case we want to change or upgrade the storage system itself - suppose we want to replace hard-disks with SSD - it should not have any impact on the logical data or schemas.

3.3 How do we transform n-ary relationships between entities in E-R diagram into relations? Give an example. (3 points)

3.4 Is a relation in 3NF also in BCNF? Justify your answer and give a relevant example. (4 points)

Part IV: Computer Networks and Data Communications (20 points)

4.1 For each of the following questions, circle the letter of the correct answer. (1 point each)

- i. A set of rules governing communication within and between computing end points is described as
 - a. Point-to-point
 - b. Protocol
 - c. Medium
 - d. Duplex
- ii. To which layer(s) of the IEEE 802 does the data link layer of the OSI/ISO reference model map?
 - a. Physical
 - b. LLC
 - c. MAC
 - d. a and b
 - e. b and c
- iii. The following are functions of a router except
 - a. Traffic isolation
 - b. Broadcast storm protection
 - c. Optimal routing
 - d. Internetworking
 - e. None of the above
- iv.. Networks in a company are best described as
 - a. Internet
 - b. internet
 - c. Intranet
 - d. Extranet
- v. Which of the following OSI/ISO layer relieves other upper layers from the burden of knowing the attached internetworking devices such as Routers, Switches, etc.?
 - a. Application layer
 - b. Data Link layer
 - c. Network Layer
 - d. Transport layer
 - e. Physical layer
- vi. In a TCP/IP peer-to-peer network, which of the following layer(s) is not required?
 - a. Data link layer
 - b. Internet/Internetworking layer
 - c. Physical layer
 - d. None of the above
 - e. All of the above
- vii. An application layer address is called
 - a. MAC address
 - b. Port number
 - c. Host name
 - d. IP address
 - e. All of the above
- viii. Which of the following layer(s) is involved in the sending and receiving hosts as well as in the intermediary nodes such as routers?
 - a. Application layer
 - b. Network layer
 - c. Transport layer
 - d. None of the above
 - e. All of the above

4.2. Mention one mechanism which is used in a data communication network to detect error during transmission _____ **Cyclic Redundancy Check (CRC)** (1 point)

the techniques used to detect noise or other impairments introduced into data while it is transmitted from source to destination.

4.3 Mention and briefly describe two media access protocols. (2 points)

4.4 What is the addressing mechanism used in Ethernet LAN? Briefly describe. (2 points)

In a LAN, **each node is assigned a physical address, also known as a MAC/Ethernet address**. This address is unique to each of the nodes on the LAN and is 6 bytes (48 bits) long, which is burned on the Ethernet card (also known as the network interface card). Ethernet is a byte-count protocol.

4.5 Mention and briefly describe at least two major reasons why IPv6 was defined to replace IPv4. (2 points)

The Internet Protocol version 6 (IPv6) is more advanced and has better features compared to IPv4. It has the capability to provide an infinite number of addresses. It is replacing IPv4 **to accommodate the growing number of networks worldwide and help solve the IP address exhaustion problem**.

4.6 In comparing circuit switching and packet switching, write **Yes** or **No** in the last two columns. (3 points)

Item	Circuit Switching	Packet Switching
Dedicated physical path required		
Each packet follows the same route		
Packets arrive in order		
A switch crash is fatal		
Potentially wasted bandwidth		
Store-and-forward transmission		

4.7 Mention and briefly describe two attenuation impairments in data transmission. (2 points)

Transmission impairment **occurs when the received signal is different from the transmitted signal**

there are three types of transmission impairments: attenuation, delay distortion, and noise. (1)

Attenuation: **The impairment is caused by the strength of signals that degrades with distance over a transmission link.**